averageInteractionsPerCrossing

MC vs data · overview · rank 30 / 98

rank 30 / 98 by endcap D_equal

used by: neither NN list
category: non-physics

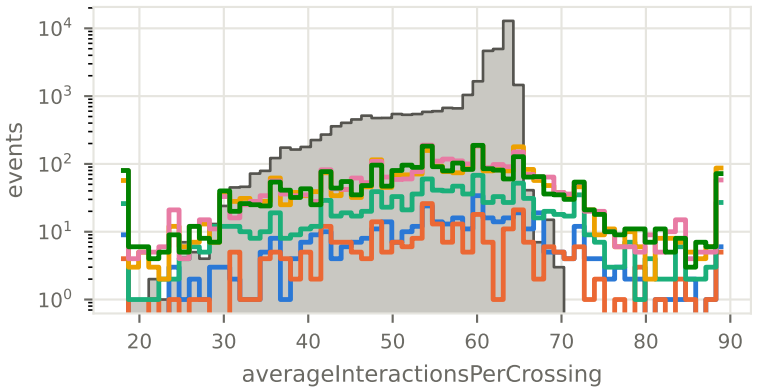
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.368	0.022	0.373	0.338	0.388	0.367	0.355	0.380	0.391	34,970	30/98
barrel	0.326	0.051	0.324	0.307	0.367	0.315	0.324	0.316	0.339	1,089	26/98
all	0.370	0.018	0.374	0.338	0.396	0.365	0.361	0.374	0.391	36,059	32/98

averageInteractionsPerCrossing

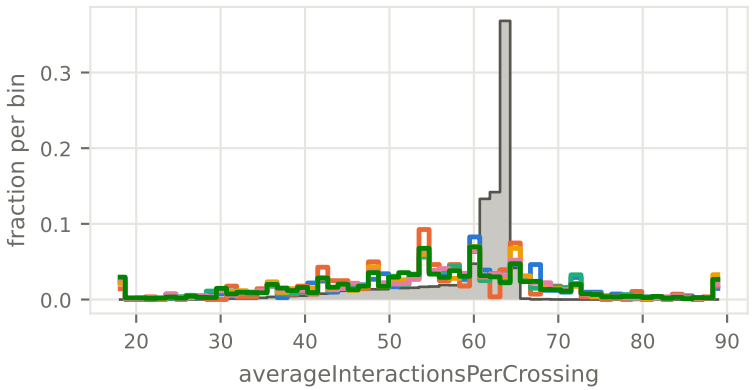
region: ENDCAP · rank 30 / 98

D_equal 0.368 (noise 0.022) D_pooled 0.373 min 0.338 (mH125_mS15) max 0.391 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

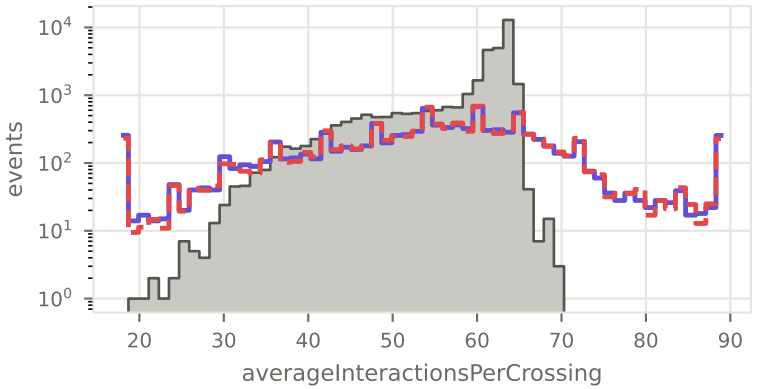


1b · each mass point vs background · normalized

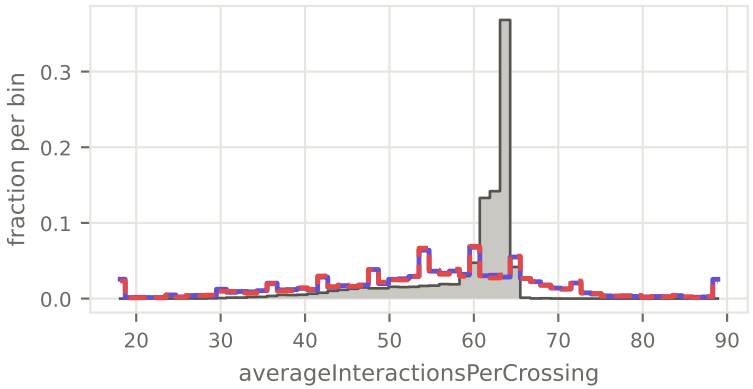


1 · each mass point
background n 34,970
mH125_mS15 D 0.338 n 411
mH125_mS55 D 0.388 n 281
mH200_mS50 D 0.367 n 1,071
mH400_mS100 D 0.355 n 2,648
mH600_mS150 D 0.380 n 2,882
mH1000_mS275 D 0.391 n 2,703

2a · combined signal vs background · raw

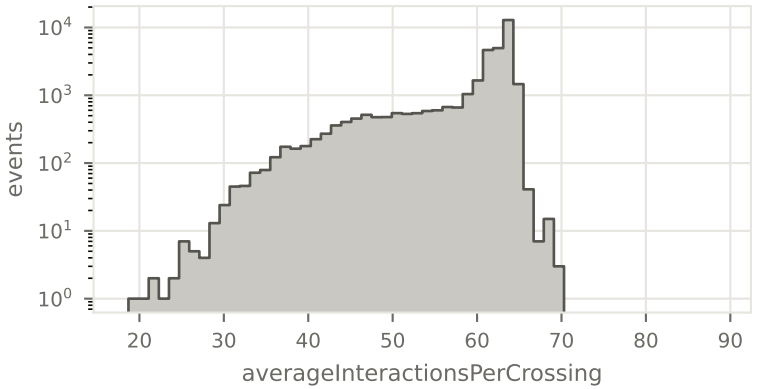


2b · combined signal vs background · normalized

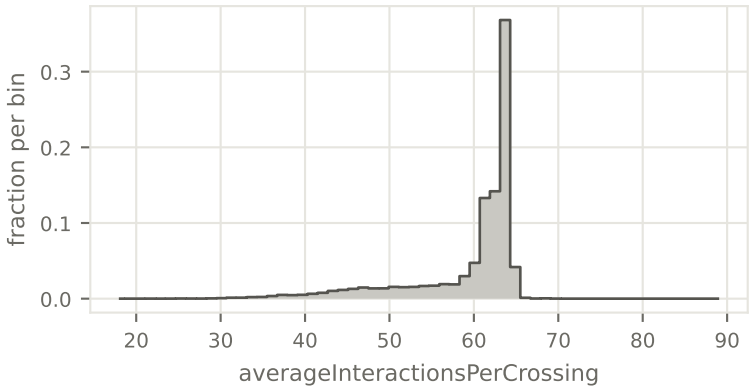


2 · combined signal
background n 34,970
combined, pooled D 0.373 n 9,996
equal weight per mass D 0.368

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

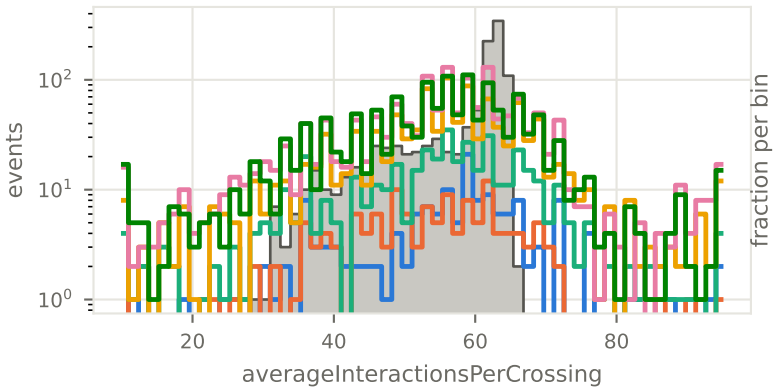
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

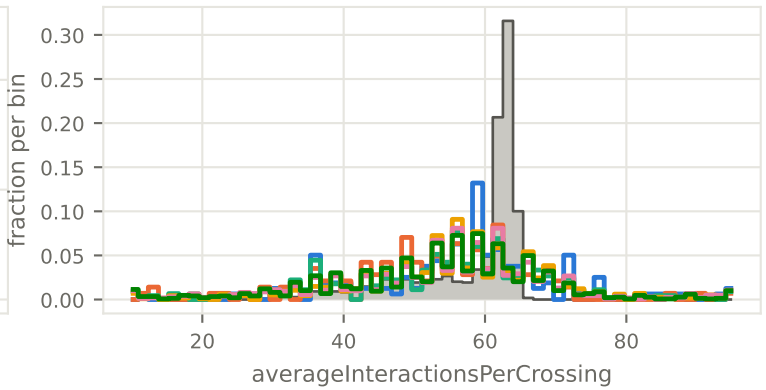
region: BARREL · rank 30 / 98

D_equal 0.326 (noise 0.051) D_pooled 0.324 min 0.307 (mH125_mS15) max 0.367 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

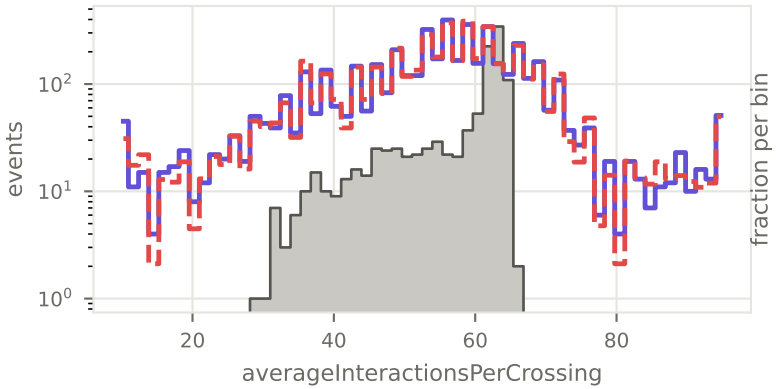


1b · each mass point vs background · normalized

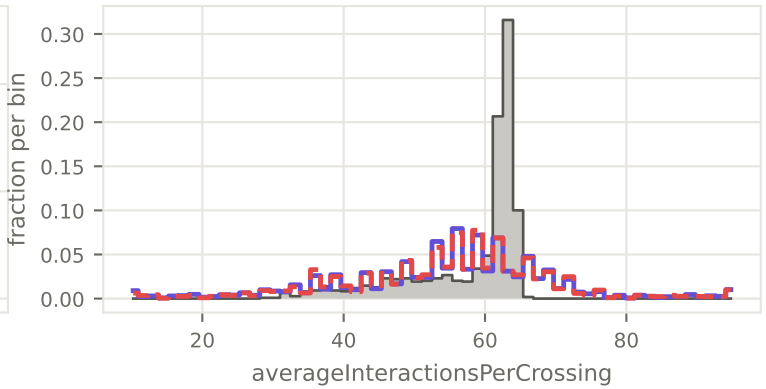


1 · each mass point
background n 1,089
mH125_mS15 D 0.307 n 159
mH125_mS55 D 0.367 n 142
mH200_mS50 D 0.315 n 449
mH400_mS100 D 0.324 n 1,144
mH600_mS150 D 0.316 n 1,608
mH1000_mS275 D 0.339 n 1,485

2a · combined signal vs background · raw

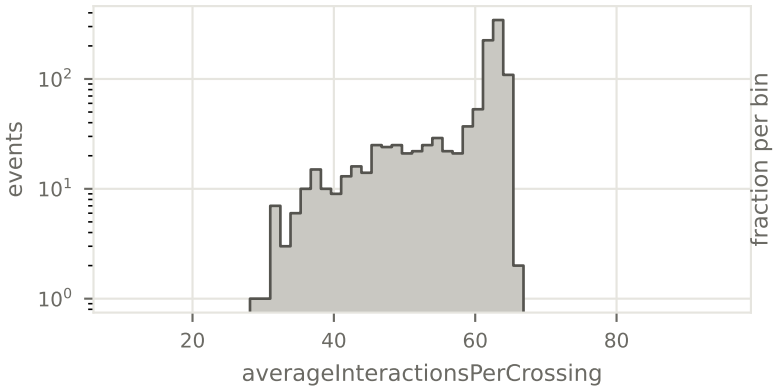


2b · combined signal vs background · normalized

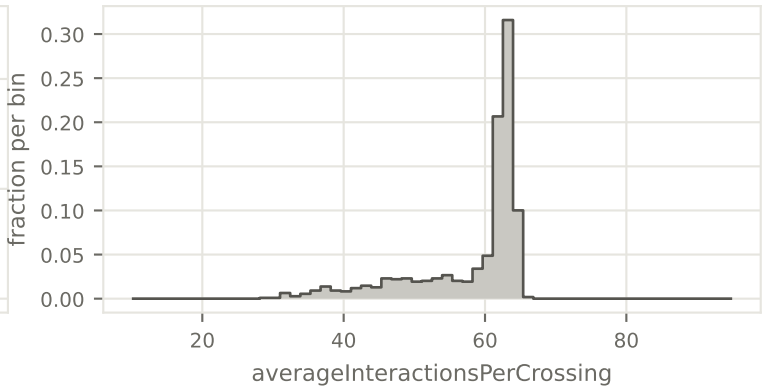


2 · combined signal
background n 1,089
combined, pooled D 0.324 n 4,987
equal weight per mass D 0.326

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

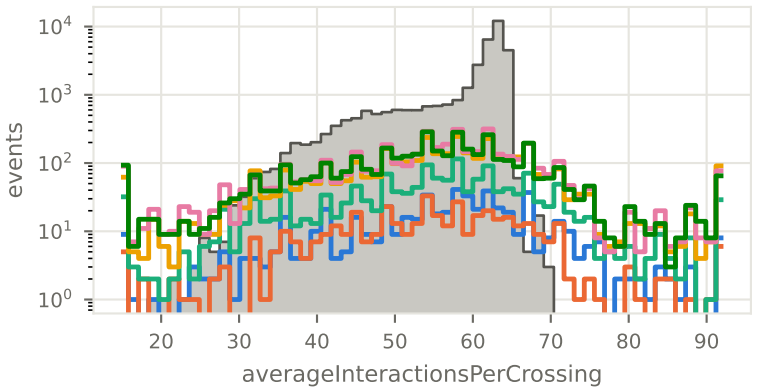
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

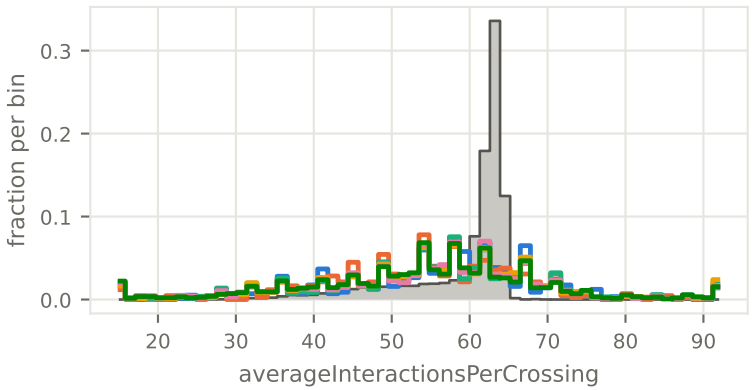
region: ALL · rank 30 / 98

D_equal 0.370 (noise 0.018) D_pooled 0.374 min 0.338 (mH125_mS15) max 0.396 (mH125_mS55) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

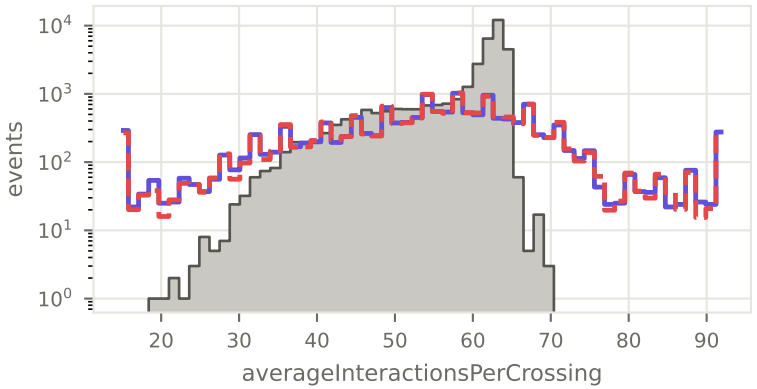


1b · each mass point vs background · normalized

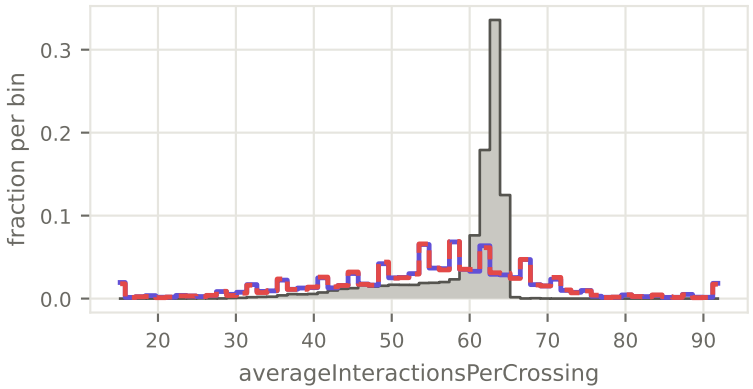


1 · each mass point
background n 36,059
mH125_mS15 D 0.338 n 570
mH125_mS55 D 0.396 n 423
mH200_mS50 D 0.365 n 1,520
mH400_mS100 D 0.361 n 3,792
mH600_mS150 D 0.374 n 4,490
mH1000_mS275 D 0.391 n 4,188

2a · combined signal vs background · raw

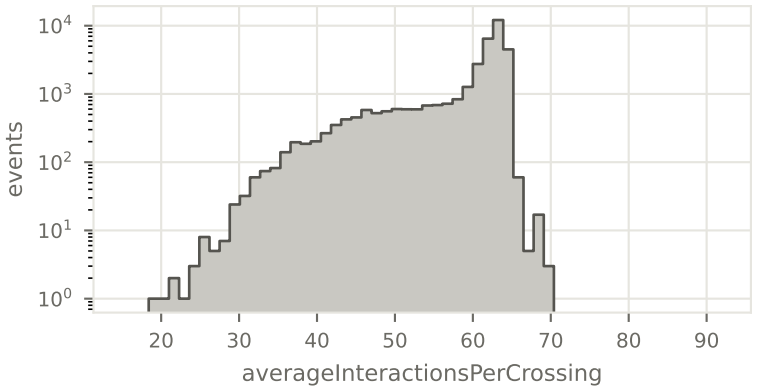


2b · combined signal vs background · normalized

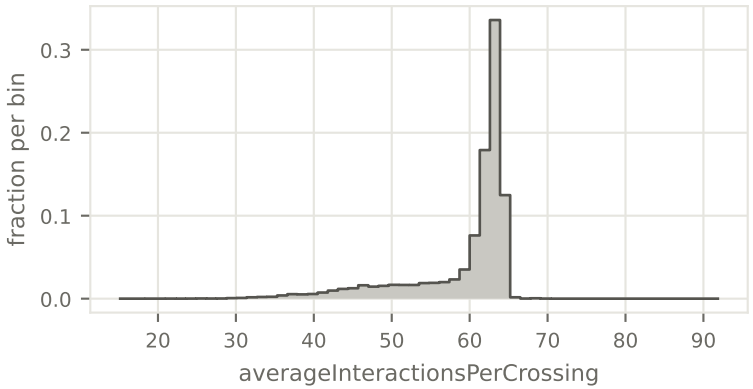


2 · combined signal
background n 36,059
combined, pooled D 0.374 n 14,983
equal weight per mass D 0.370

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed